Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Net force

Meaning in my words How would you explain net force to a friend?	**How can you tell?** Two people push one cart in opposite directions. Why do the arrow directions matter?
Example or sketch One person pushes 6 N right and another 4 N left. What single arrow replaces both?	**Non-example and why** A student adds those pushes to get 10 N right. Which push did they count the wrong way?

Acceleration

Meaning in my words How would you explain acceleration to a friend?	**How can you tell?** A cart follows a bend at steady speed. Why is it accelerating?
Example or sketch Positive is right. A cart goes from −2 to −6 m/s in 2 s. What acceleration fits?	**Non-example and why** A cart travels straight at −6 m/s the whole time. Does the minus sign mean it is accelerating?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Velocity

Meaning in my words How would you explain velocity to a friend?	**How can you tell?** Right is positive. What does a velocity of −2 m/s tell you?
Example or sketch How would you show a cart going east at 3 m/s?	**Non-example and why** A speedometer reads 3 m/s. What would you still need for velocity?

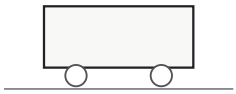
Speed trend

Meaning in my words How would you explain speed trend to a friend?	**How can you tell?** A cart moves left but accelerates right. What happens to its speed before it stops?
Example or sketch Velocities one second apart are −6, −4 and −2 m/s. Describe the trend.	**Non-example and why** A student says −2 to −6 m/s means the cart slowed. Which numbers should they compare?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw an object with two opposing forces. Write the signed sum under the net arrow.



My marks mean

On assignment: Net force

SECTION E / BACK / FINAL MEETING ONLY Name _____

A tracking model moves left. Its 4 kg body feels 12 N right and 4 N left. The status display has separate motion and force arrows.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Controls engineer**

Set the net-force and acceleration arrows. Tell the operator what happens to speed at first while velocity points left.

☐ **Replay analyst**

Annotate the slow-motion clip with one arrow for motion and another for acceleration. Explain why they can point opposite ways.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor's check: Say what information is needed to predict when it reverses.